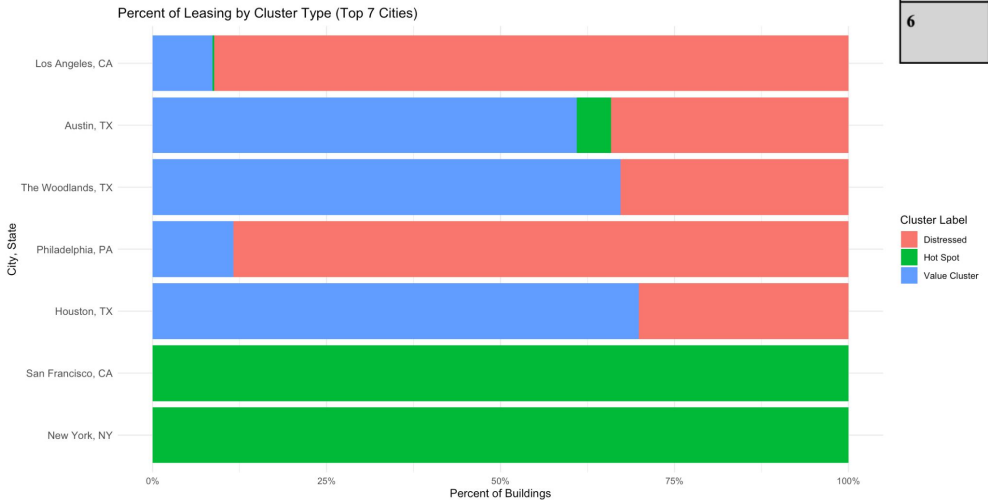
<u>Objective</u>: Specify **where** and **when** to lease offices based on **geographic clusters** and **industry sectors**.

GENERAL FRAMEWORK

The Elbow Method → # of clusters
Label assigned based on characteristics

cluster	label	avg_rent	avg_occupancy	avg_leasedSF	count	Properties
1	Transitional large leases	64.45989	0.3982879 (moderate)	129529.421 (high)	466	Signs of big deals
2	Hot spot	69.49608 (high)	0.4725546 (good)	8467.758	8263	Premium market with strong demand
3	Distressed	39.22149 (low)	0.3733413 (low)	4642.829 (low)	15399	Underperforming/Challenged
4	Value Cluster	30.16013 (low)	0.6081863 (high)	4274.574	9335	Affordable but in-demand => Long term investment
5	Overpriced/Risky	69.18995 (high)	0.2053649 (low)	8349.132	4270	Possibly inflated prices
6	Mega Leases	66.21839	0.4259491	478491.681 (high)	47 (very high)	Large tenants (HQs, data centers,..)



- Top **7** cities with the highest lease.sf are considered
- Each city is grouped based on **prevalent clusters**

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                 Financial Services and Insurance Technology, Advertising, Media, and Information
Distressed                                   188                                              193
Hot Spot                                     663                                              429
Value Cluster                                 66                                               80
                 Financial Services and Insurance Technology, Advertising, Media, and Information
Distressed                                -1.892301                                         2.162750
Hot Spot                                   1.789007                                        -2.044693
Value Cluster                             -1.835817                                         2.098193
$Chi_Square_Test

        Pearson's Chi-squared test

data:  contingency_table
X-squared = 23.412, df = 2, p-value = 8.243e-06


$Standardized_Residuals
                 Financial Services and Insurance Technology, Advertising, Media, and Information
Distressed                                -2.873724                                         2.873724
Hot Spot                                   2.716858                                        -2.716858
Value Cluster                             -2.787945                                         2.787945
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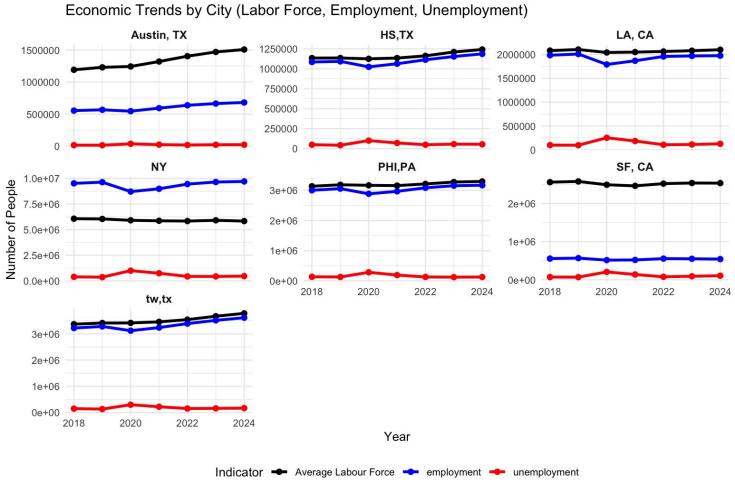


Chi-test: **Proven correlation** between cluster type and the two sectors

WHERE TO MOVE IN

City	Cluster Type	Economic Trends	Dominant Industry
New York, NY	Hot Spot	High labor force, improving unemployment	Finance-heavy
San Francisco, CA		Stable labor force, employment rebounding	Tech-heavy
Houston, TX	Value Cluster	Rebounding labor/employment, stable outlook	Tech-heavy
The Woodlands, TX		High labor growth, low unemployment	
Austin, TX		Fast-growing employment and labor force	
Los Angeles, CA	Distressed	Slow growth, elevated unemployment	Tech-heavy
Philadelphia, PA		Weak recovery, consistently high unemployment	Finance-heavy

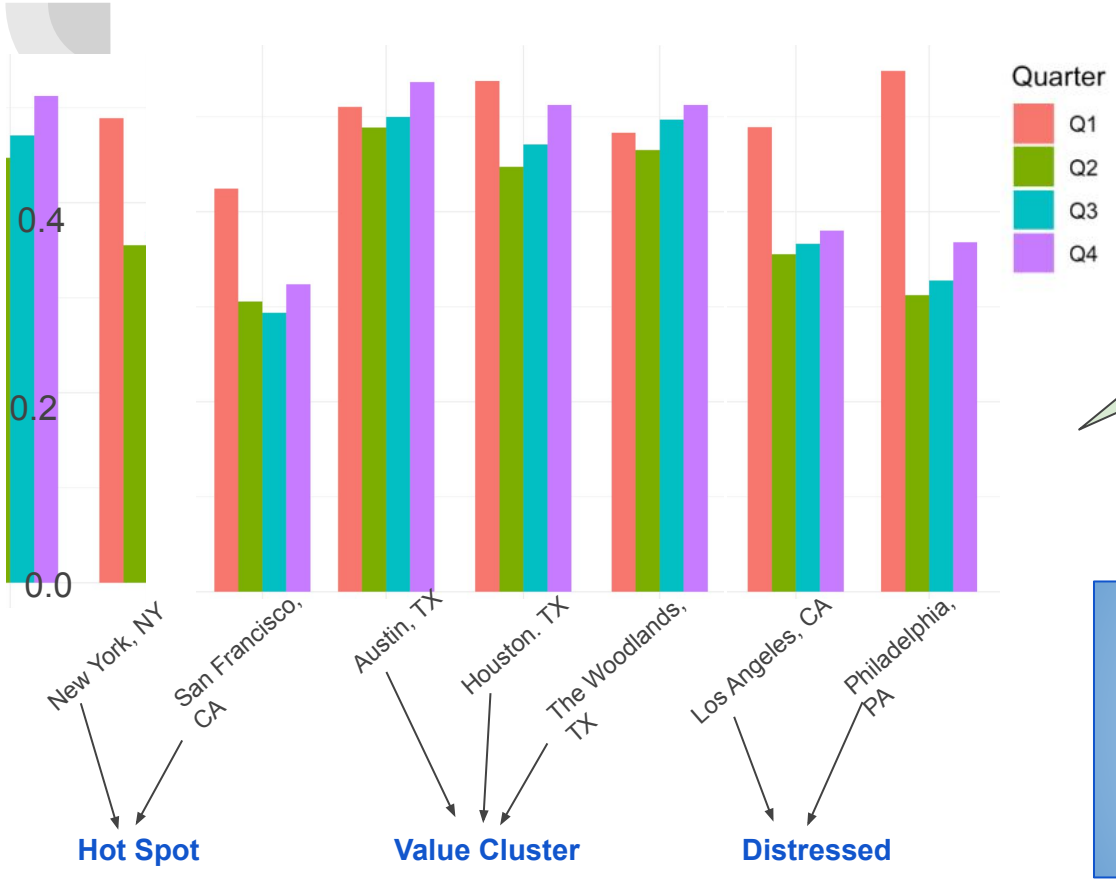
Average labor force, employment and unemployment rates help firms to see:

- Pools of labor
- Economic growth

=> Tailored to firm's conditions and needs (startup, budget-conscious,..)

WHEN TO MOVE IN

Average Occupancy Proportion Per Qu



Lower occupancy proportions were often observed in Q2 and Q3

H0: $(\mu1+\mu4)/2 = (\mu2+\mu3)/2$
HA: $(\mu1+\mu4)/2 \neq (\mu2+\mu3)/2$
t-value = 2.1658
p_value = **0.0397** < 0.005
→ moderate evidence